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Engineering

Frontiers of Graph Neural Networks with DIG

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Department of Computer Science & Engineering

Schedule



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Generation	dig.ggraph	35min
Break		10min
Self-supervised learning	dig.sslgraph	35min
Break		20min
Explainability	dig.xgraph	35min
Break		10min
3D graphs	dig.3dgraph	35min



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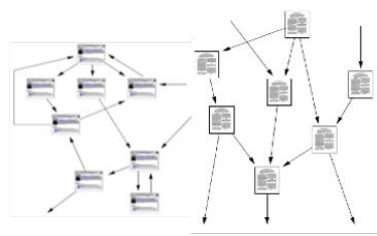
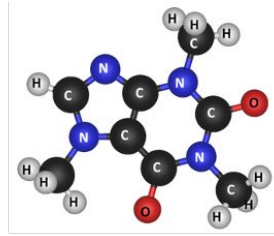
Background Knowledge

Graphs



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Objects + Relationships = Graph



.....

Tasks over Graphs



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- Node-level classification/regression
- Graph-level classification/regression
- Link prediction
- Generation
- Explainability
-



The representation of each node is updated iteratively by aggregating the representations of its neighbors.

$$\mathbf{a}_v^{(\ell)} = f_{\text{aggregate}}^{(\ell)} \left(\{ \{ \mathbf{h}_u^{(\ell-1)} \mid u \in \mathcal{N}_v \} \} \right)$$

$$\mathbf{h}_v^{(\ell)} = f_{\text{update}}^{(\ell)} \left(\mathbf{h}_v^{(\ell-1)}, \mathbf{a}_v^{(\ell)} \right)$$



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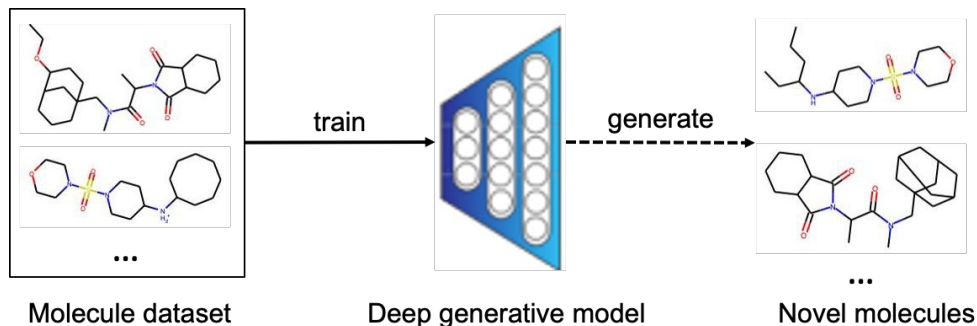
Graph Generation

Problem



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- Train generative models to model the data distribution (explicitly/implicitly).



Challenges



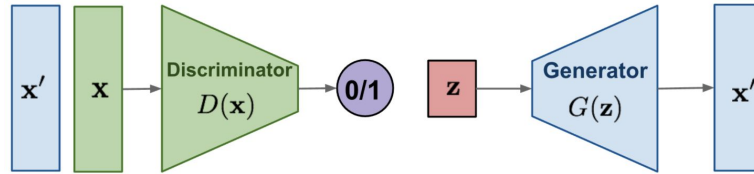
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- Discrete data
- Permutation invariance
- Non-grid data: various size and topology
- Evaluation

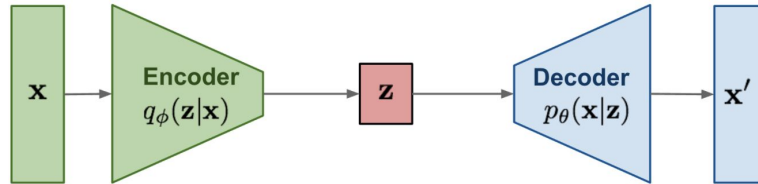
Generative Models



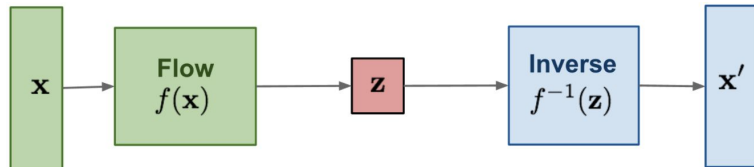
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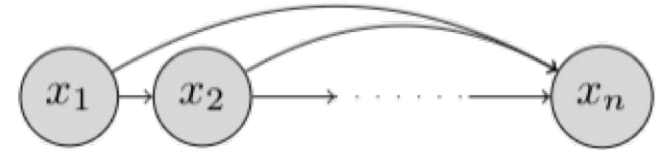
GAN



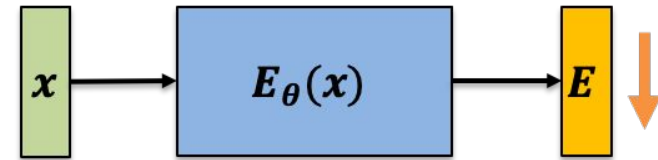
VAE



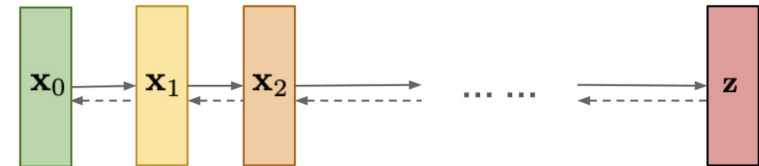
Flow models



Deep autoregressive models



EBM



Diffusion models

Figures are from <https://lilianweng.github.io/>, <https://deepgenerativemodels.github.io/>.

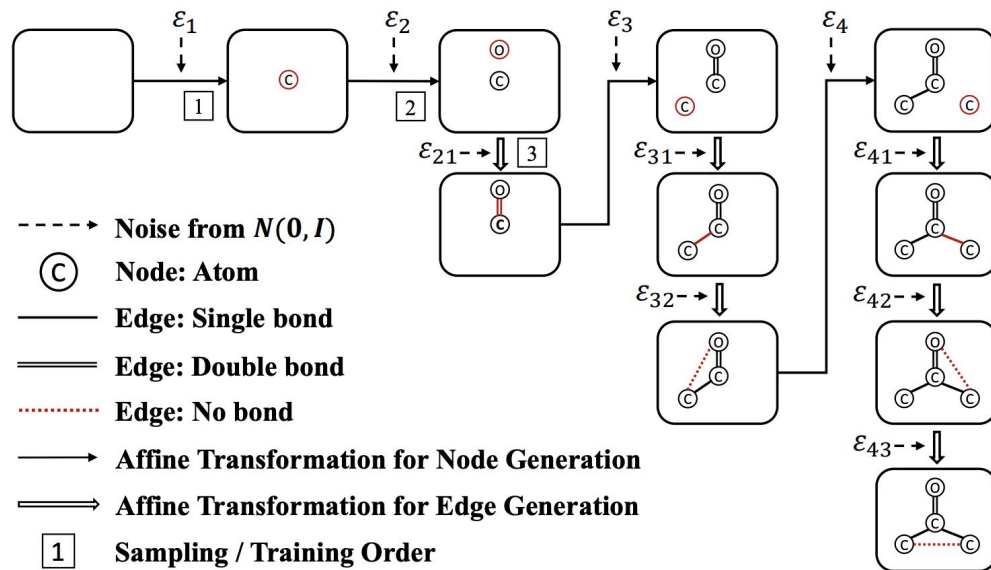
Graph Generation: Overview



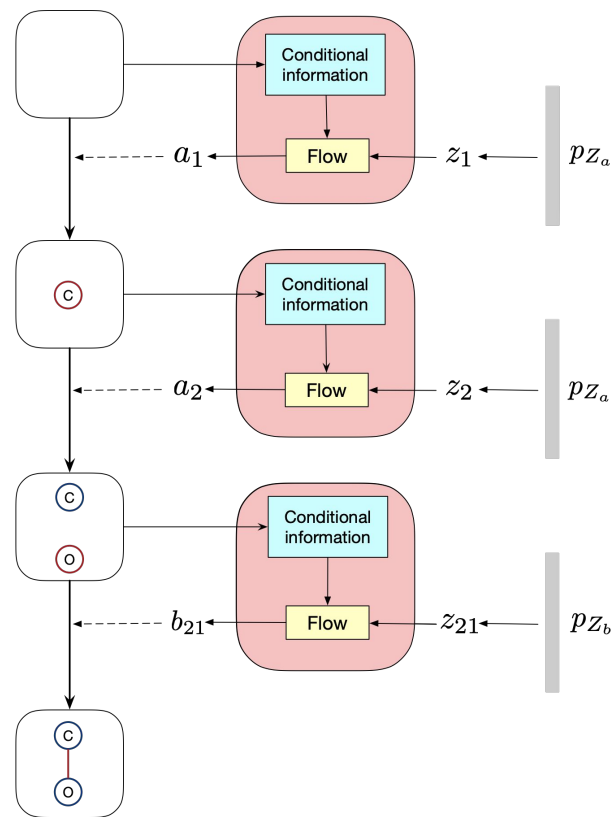
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	Atom-by-atom	Motif-by-motif	One-shot
GAN	GCPN		MolGAN
VAE	CGVAE, JT-VAE	HierVAE	GraphVAE, RVAE
Flow models	GraphAF, GraphDF		GraphNVP, GRF, MoFlow, GraphCNF
Deep autoregressive models	DeepGMG, MolecularRNN		
EBM			GraphEBM
Diffusion models			GDSS

- Decompose the molecular graph into a sequence of continuous tokens
 - Dequantization
 - Autoregressive flow models
 - Affine transformation
-
- Not discrete
 - Not permutation-invariant

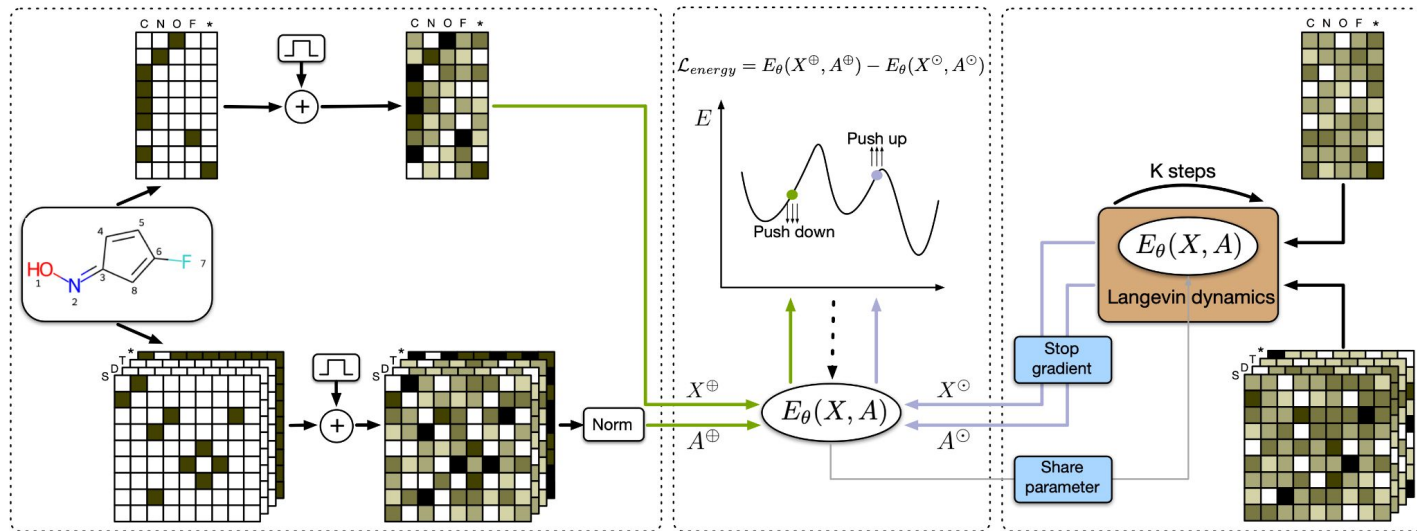


- Decompose the molecular graph into a sequence of **discrete** tokens
- Discrete autoregressive flow models
 - Invertible modulo shift transformation
- **Not permutation-invariant**

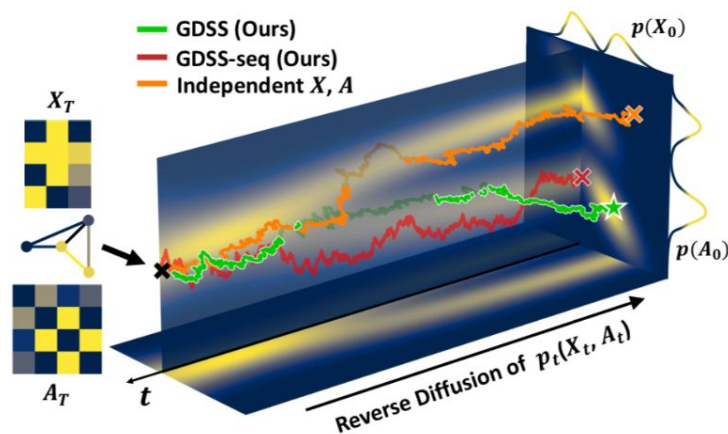


- Energy-based models: **permutation-invariant**
- Multi-objective generation via compositional generation

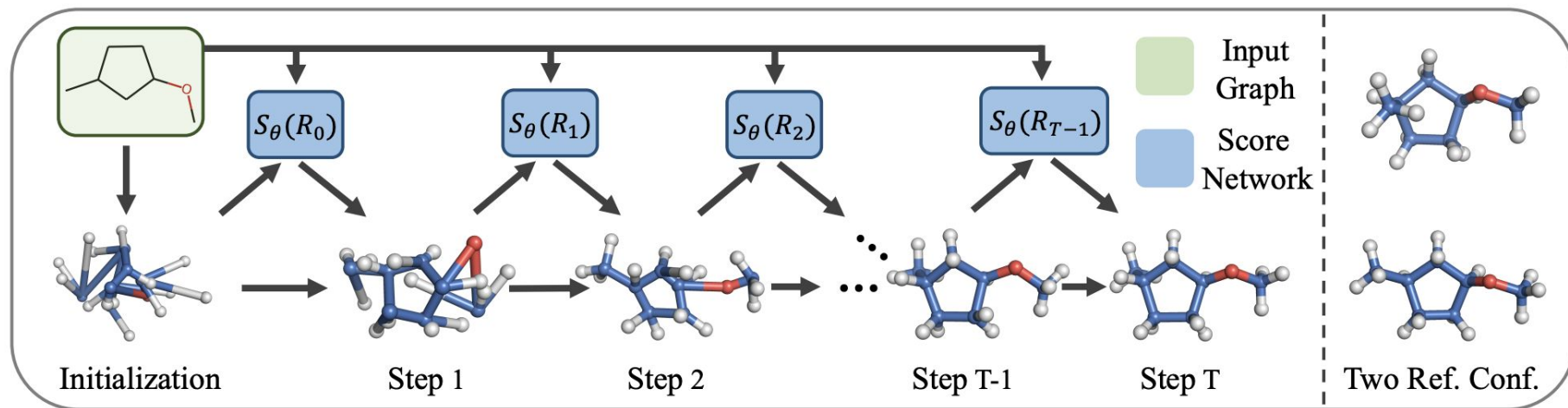
- **Not discrete**



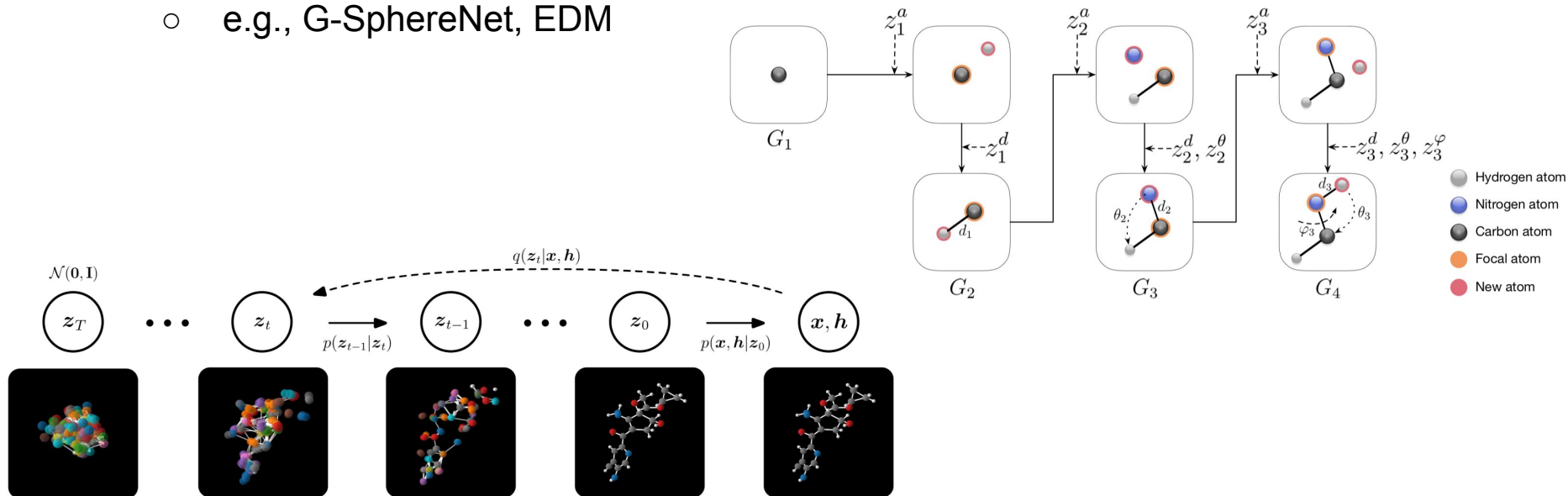
- Diffusion models: **permutation-invariant**
- Model the training distribution well
- **Not discrete**, but empirically found that the entries of the resulting samples after the simulation of the diffusion process do not deviate much from the integer values 0 and 1.



- Molecular geometry generation from 2D graph
 - e.g., ConfGF



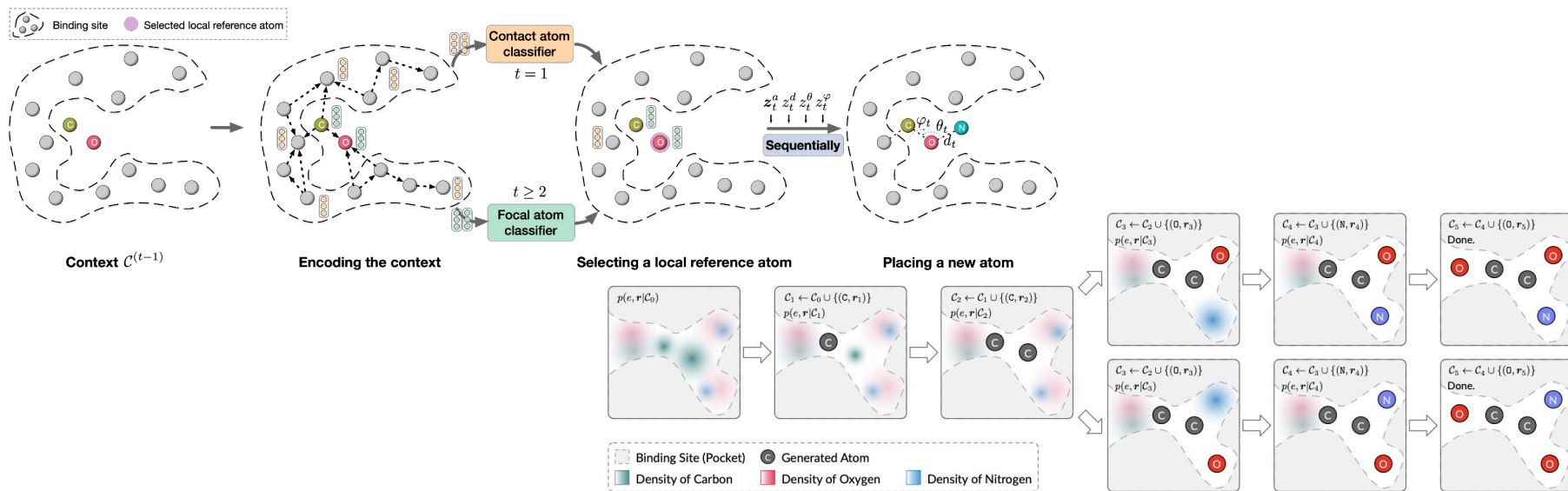
- Molecular geometry generation from scratch
 - e.g., G-SphereNet, EDM



Luo et al. An Autoregressive Flow Model for 3D Molecular Geometry Generation from Scratch. ICLR, 2022.

Hoogeboom et al. Equivariant Diffusion for Molecule Generation in 3D. ICML, 2022.

- Molecular generation for target protein: structure-based drug design
 - e.g., Luo et al., GraphBP



Luo et al. A 3D Generative Model for Structure-Based Drug Design. NeurIPS, 2021.

Liu et al. Generating 3D Molecules for Target Protein Binding. ICML, 2022.



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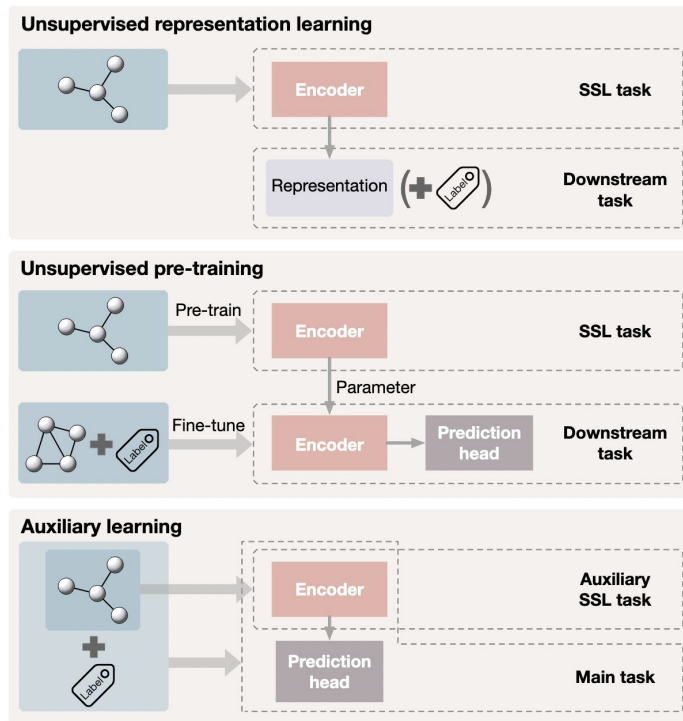
Self-Supervised Learning on Graphs

Self-supervised Learning



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- Training models with **self-supervision**.
- Success in text and image data. We see an explosion of graph SSL papers.
- Three paradigms of using self-supervised learning
 - Self-supervised representation learning
 - Unsupervised pre-training
 - Auxiliary learning



Xie et al. Self-Supervised Learning of Graph Neural Networks: A Unified Review. TPAMI, 2022.

Taxonomy of SSL Methods



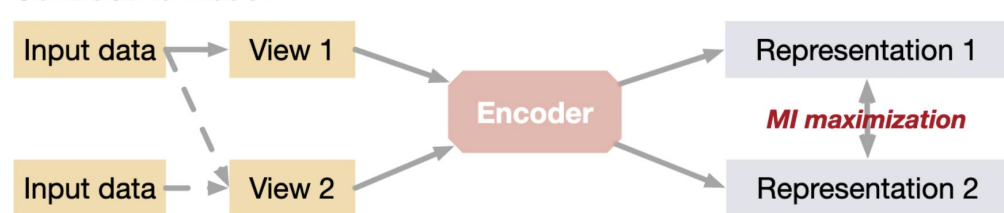
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Contrastive vs Predictive: are negative pairs required?

- Contrastive

- Maximizes mutual information
- Focuses on designing view generation

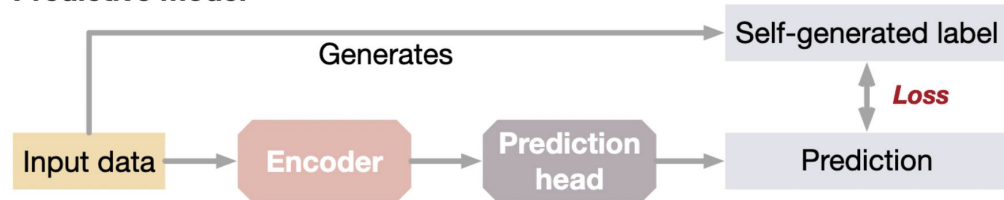
Contrastive model



- Predictive

- Ad-hoc pretext-tasks / information bottleneck
- Focuses on designing task/objective

Predictive model

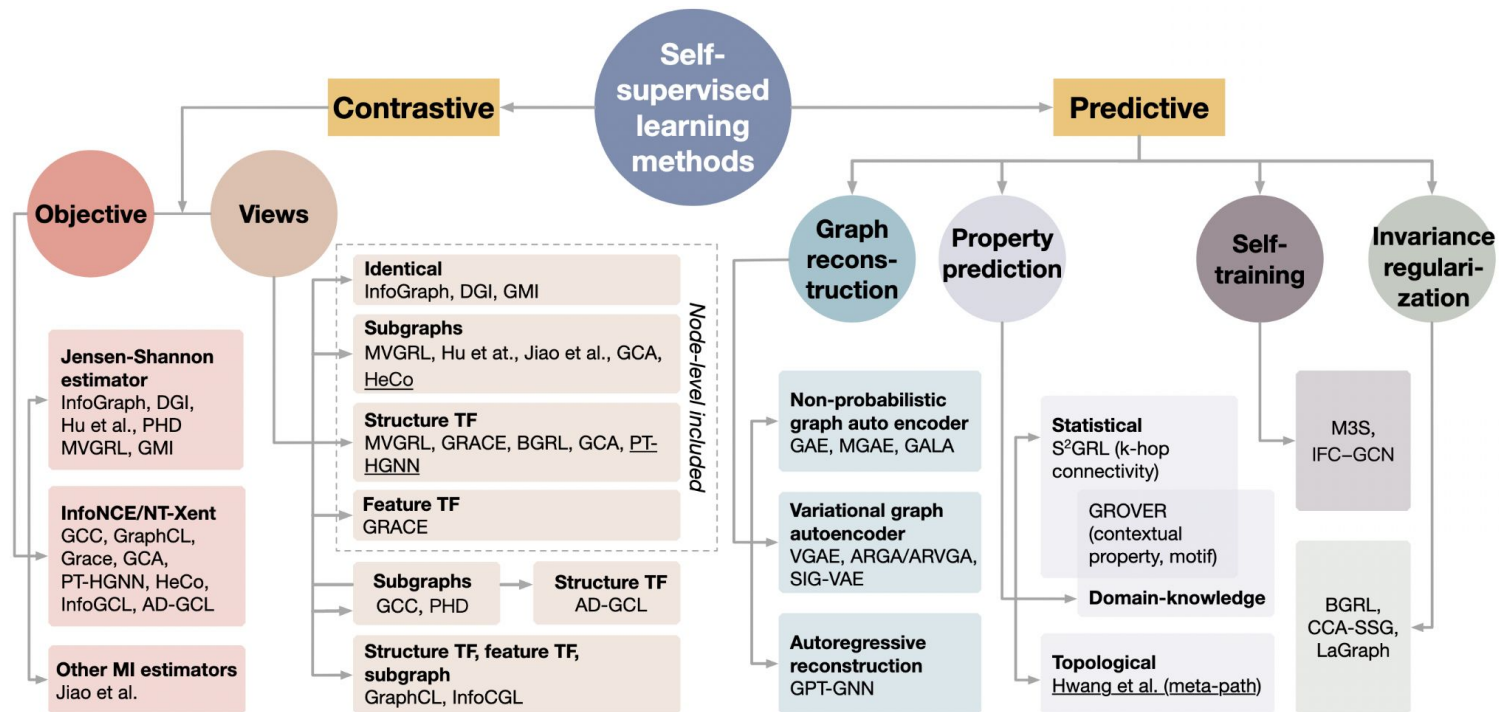


Xie et al. Self-Supervised Learning of Graph Neural Networks: A Unified Review. TPAMI, 2022.

Taxonomy of SSL Methods



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Xie et al. Self-Supervised Learning of Graph Neural Networks: A Unified Review. TPAMI, 2022.

- Representations of “similar” graph data to be closer, and that of “dissimilar” graph data to be further from each other.
- In practice, we do not have ground-truth or defined measurement for similarity. We construct augmented graphs (or subgraphs) from the same graph to be “similar”, and different graph samples are “dissimilar”.

- Representations of “similar” graph data to be closer, and that of “dissimilar” graph data to be further from each other.
- Theoretical grounding: mutual information maximization. Jointly sampled graphs are “similar”, while independently sampled graphs are “dissimilar”

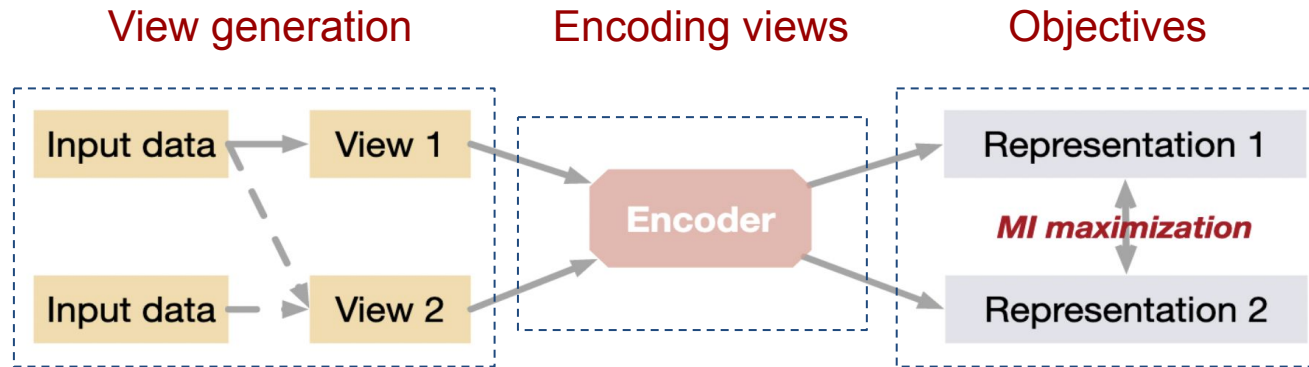
$$\mathcal{I}(\mathbf{x}, \mathbf{y}) = D_{KL}(p(\mathbf{x}, \mathbf{y}) || p(\mathbf{x})p(\mathbf{y}))$$

- We consider a graph as a random variable and obtain two views of the graph.
- Representations of the two views are $\mathbf{x}$ and $\mathbf{y}$. A good GNN should encode two views into representation that share as much mutual information as possible.

Contrastive Methods



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Xie et al. *Self-Supervised Learning of Graph Neural Networks: A Unified Review*. TPAMI, 2022.

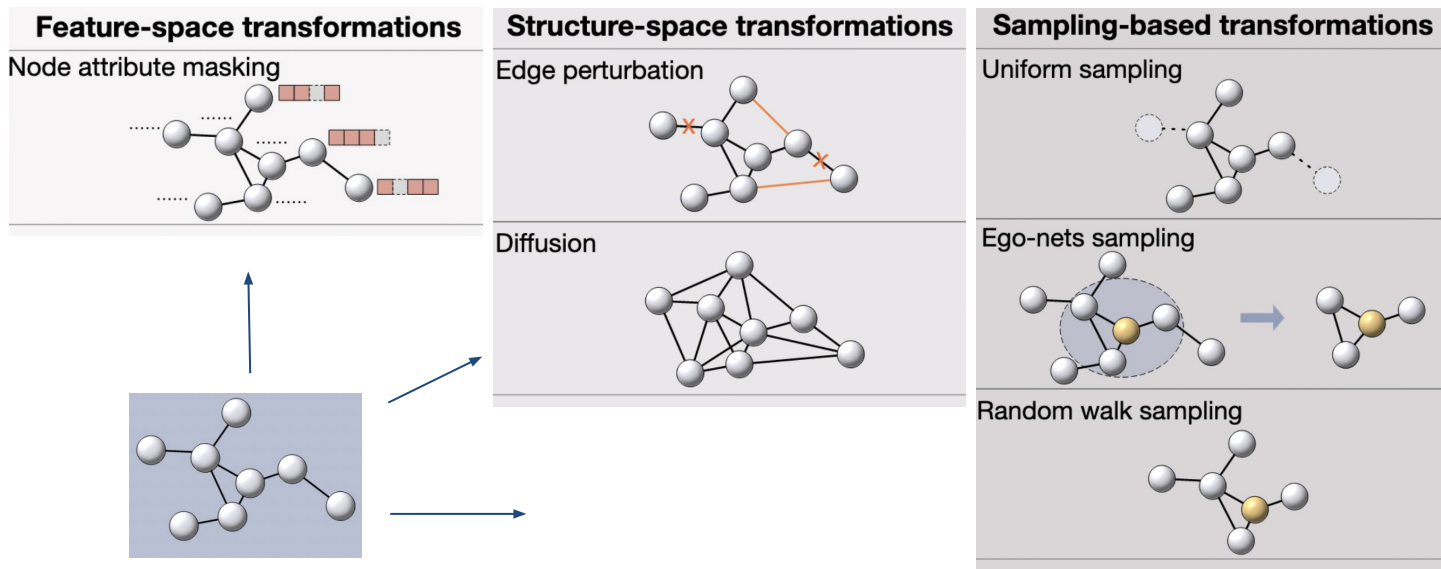
- Derived as lower bounds of mutual information
 - Jensen-Shannon Estimator (DGI, InfoGraph, MVGRL, etc.)

$$\hat{\mathcal{I}}^{(JS)}(\mathbf{h}_i, \mathbf{h}_j) = \mathbb{E}_{(\mathbf{A}, \mathbf{X}) \sim \mathcal{P}} [\log(\mathcal{D}(\mathbf{h}_i, \mathbf{h}_j))] + \mathbb{E}_{[(\mathbf{A}, \mathbf{X}), (\mathbf{A}', \mathbf{X}') \sim \mathcal{P} \times \mathcal{P}]} [\log(1 - \mathcal{D}(\mathbf{h}_i, \mathbf{h}'_j))]$$

- InfoNCE (GraphCL, GRACE, etc.)

$$\begin{aligned} \hat{\mathcal{I}}^{(\text{NCE})}(\mathbf{h}_i, \mathbf{h}_j) &= \mathbb{E}_{(\mathbf{A}, \mathbf{X}) \sim \mathcal{P}} \left[\mathcal{D}(\mathbf{h}_i, \mathbf{h}_j) - \mathbb{E}_{\mathbf{K} \sim \mathcal{P}^N} \left[\log \sum_{(\mathbf{A}', \mathbf{X}') \in \mathbf{K}} e^{\mathcal{D}(\mathbf{h}_i, \mathbf{h}'_j)} / N \mid (\mathbf{A}, \mathbf{X}) \right] \right] \\ &= \mathbb{E}_{[(\mathbf{A}, \mathbf{X}), \mathbf{K}] \sim \mathcal{P} \times \mathcal{P}^N} \left[\log \frac{e^{\mathcal{D}(\mathbf{h}_i, \mathbf{h}_j)}}{\sum_{(\mathbf{A}', \mathbf{X}') \in \mathbf{K}} e^{\mathcal{D}(\mathbf{h}_i, \mathbf{h}'_j)}} \right] + \log N \end{aligned}$$

- Three types of graph transformations for view generation

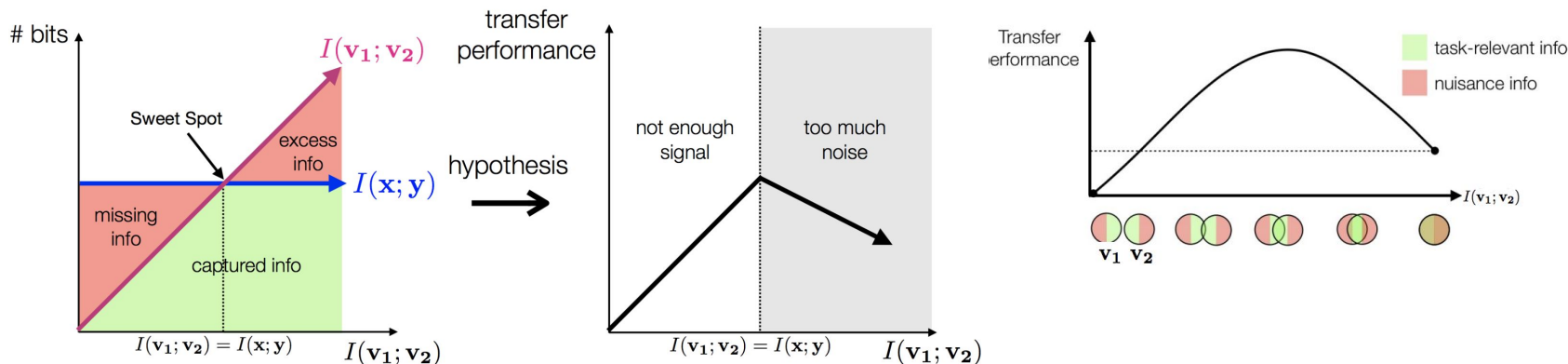


What are good views for graphs?



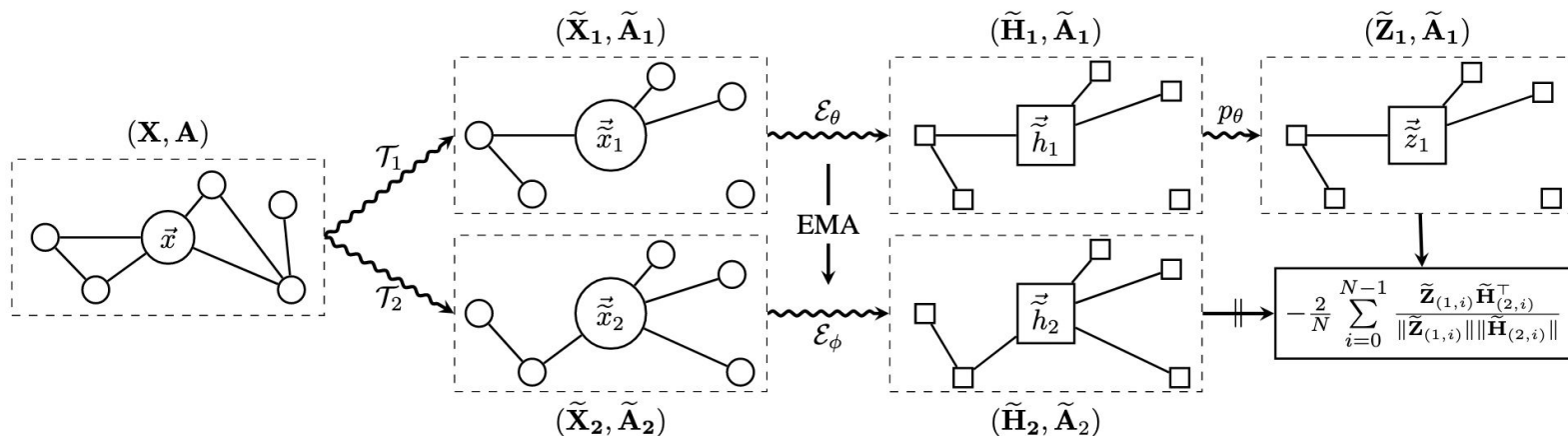
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- Ideally, good views (v_1, v_2) generated for contrastive learning should
 - Have their mutual information $I(v_1, v_2)$ minimized
 - Subject to $I(v_1, y) = I(v_2, y) = I(x, y)$



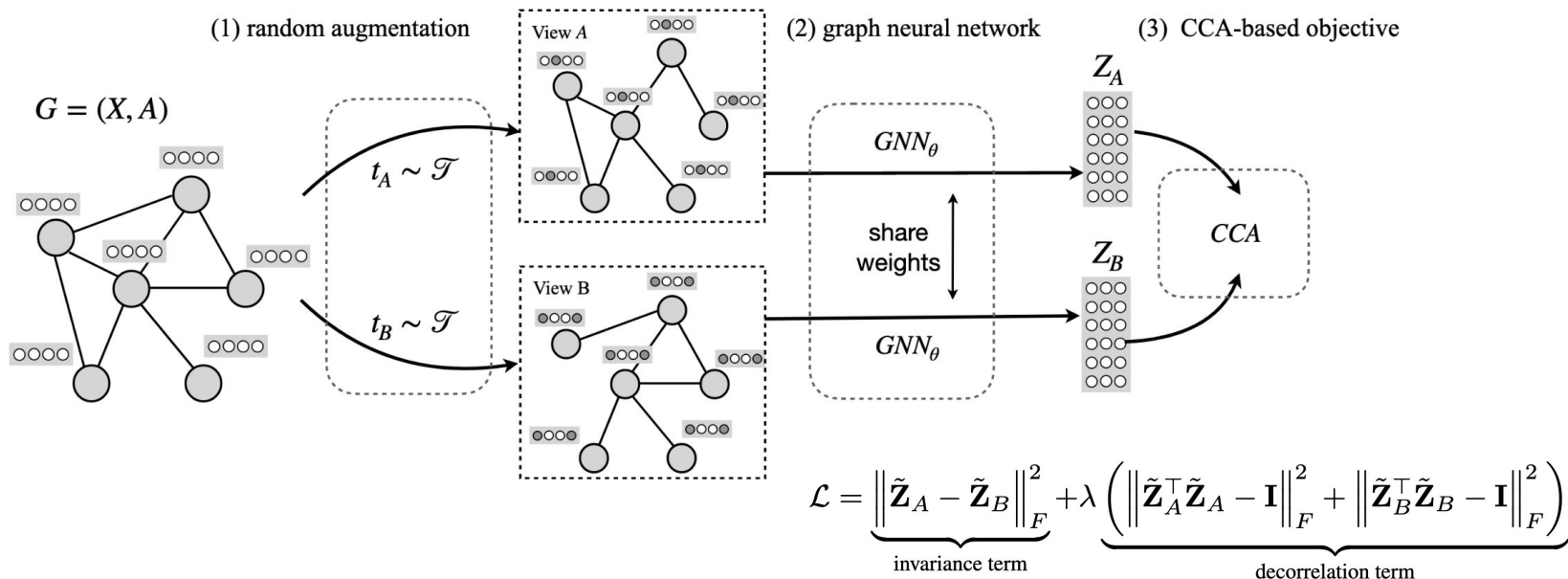
- The principle is even **more important for graphs.**

- BGRL

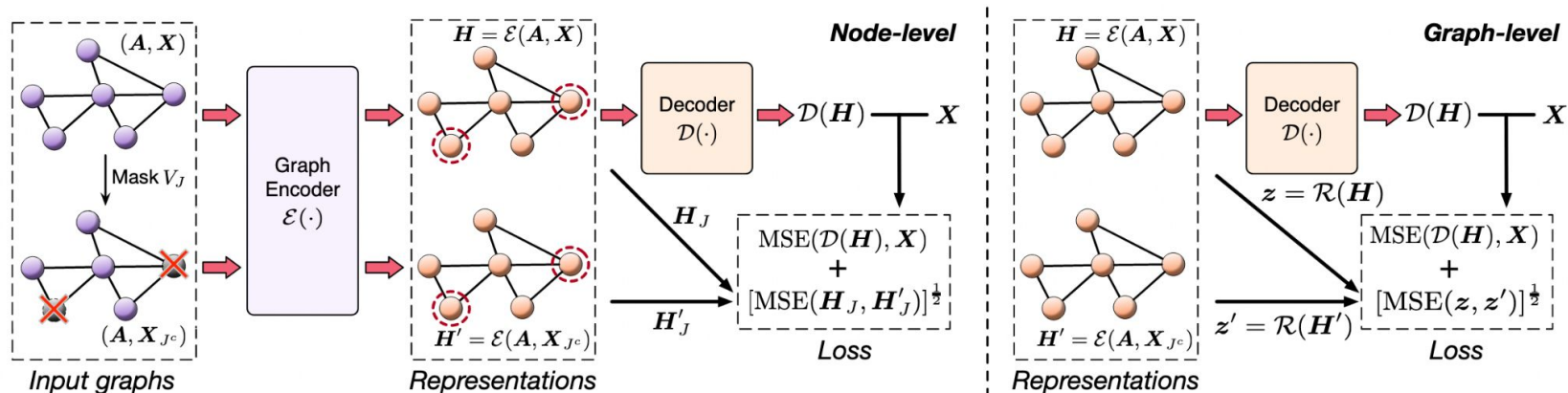


$$\theta \leftarrow \text{optimize}(\theta, \eta, \partial_\theta \ell(\theta, \phi)) \quad \text{EMA update: } \phi \leftarrow \tau \phi + (1 - \tau) \theta$$

- CCA-SSG

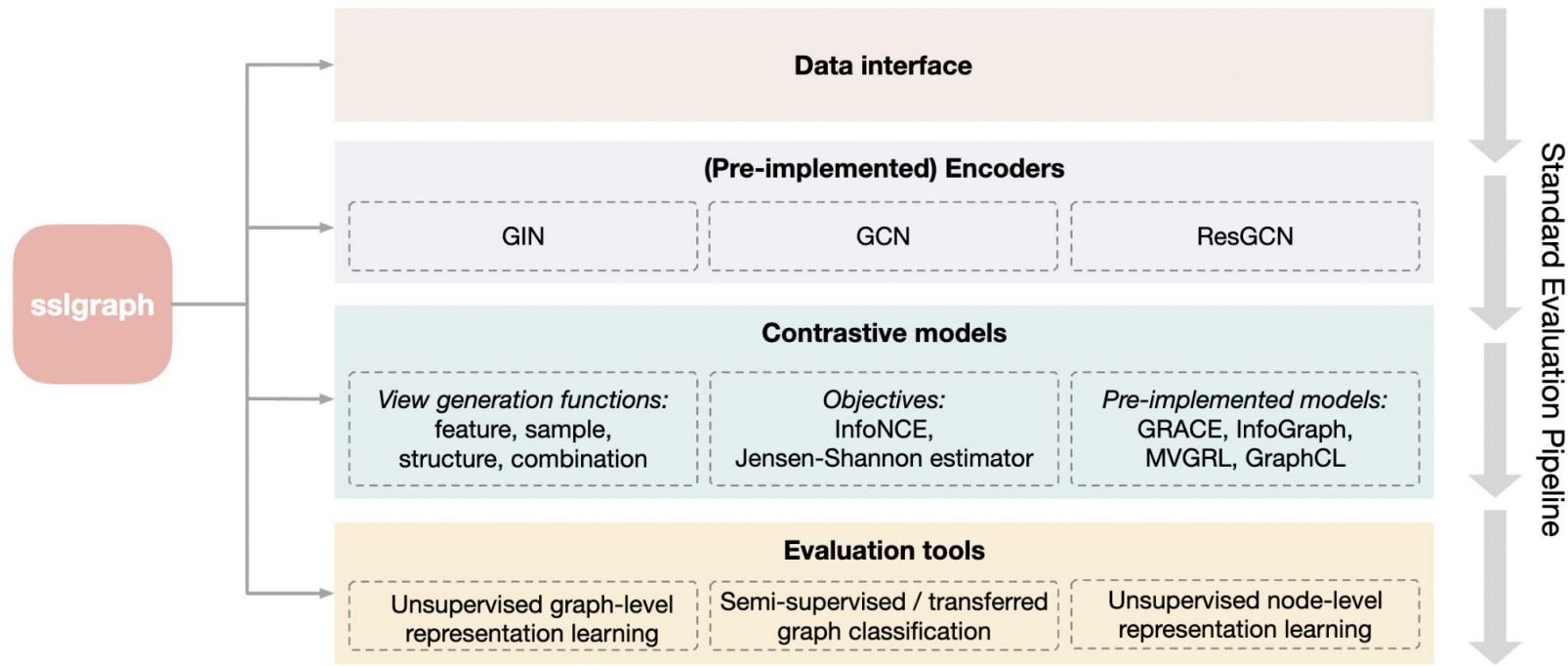


- LaGraph



$$L_{node}(\mathcal{E}, \mathcal{D}) = \frac{1}{N} \sum_{i=1}^N \|\mathcal{D}(A_i, H_i) - X_i\|^2 + \alpha \left[\frac{\sum_i \|\mathbb{1}_{J_i} \odot H_i - \mathbb{1}_{J_i} \odot H'_i\|^2}{\sum_i |J_i|} \right]^{1/2} \quad \Bigg| \quad + \alpha' \left[\frac{\sum_i \|z_i - z'_i\|^2}{\sum_i |J_i|} \right]^{1/2}$$

- Similarity
 - Do not requires negative samples.
 - Better scalability.
 - Their objectives include a term to regularize the invariance of representations to different views (minimize their distance)
 - Effectiveness can be justified by the **Information Bottleneck Principle**
- Difference
 - The formulation of entire objectives are different.
 - Based on or derived from different theoretical grounding/insights
 - Different technique to guarantee informative representations





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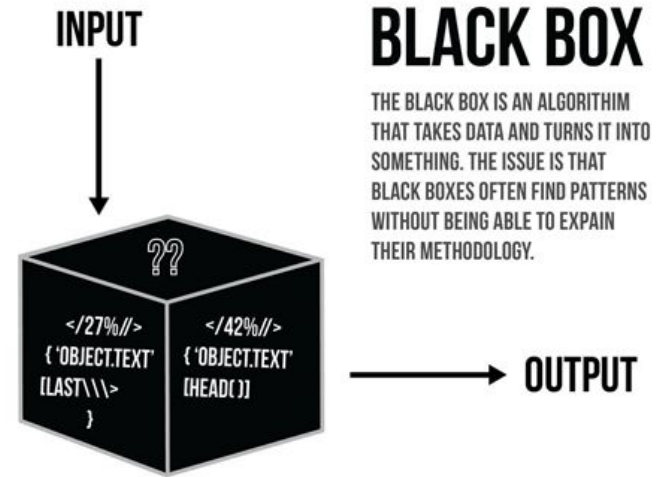
Explainability of GNNs

Why Explainability



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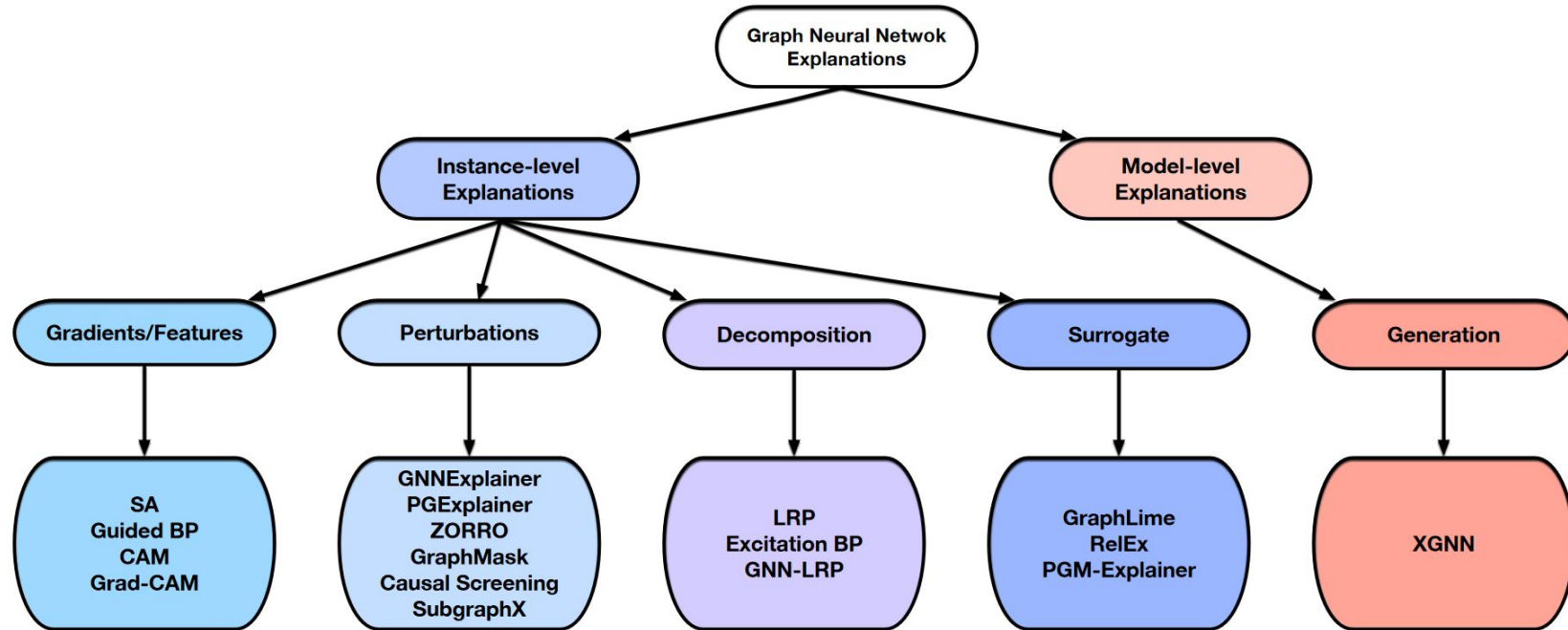
- People don't understand deep model
- People don't trust deep model



Reference: <https://towardsdatascience.com/guide-to-interpretable-machine-learning-d40e8a64b6cf>

- What's the explanation level
 - Instance-level: edge-based, subgraph-based
 - Model-level: xgmn
- How to verify the explanation results are meaningful to human
 - Graph Sentiment dataset
- Hard to compare explainability methods systematically
 - Quantitative metrics: Fidelity+, Fidelity-, Sparsity

Overview of methods



Explanation on GNNs

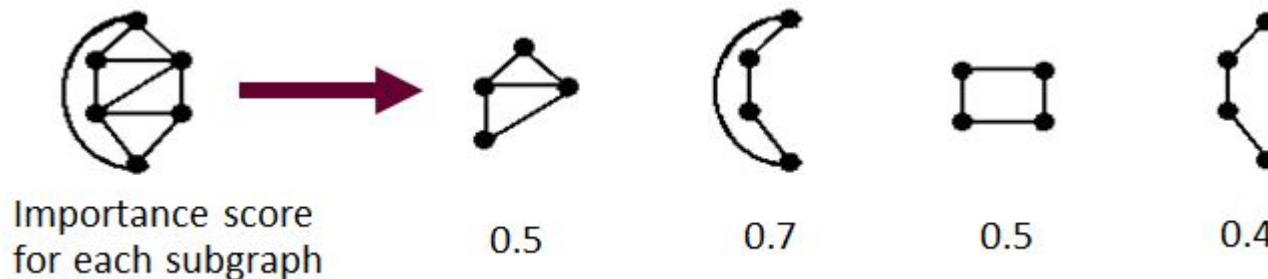


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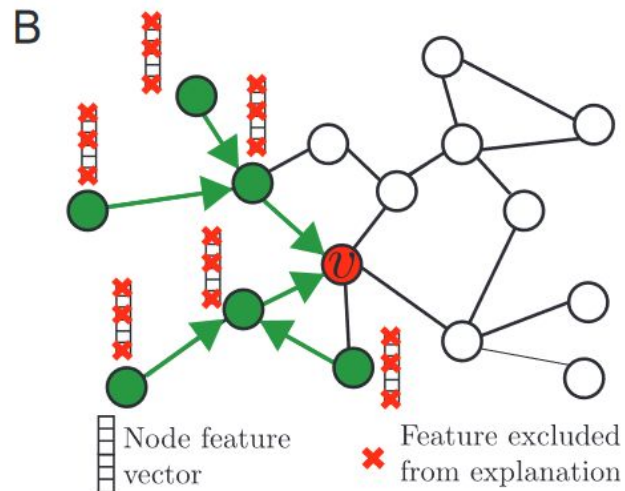
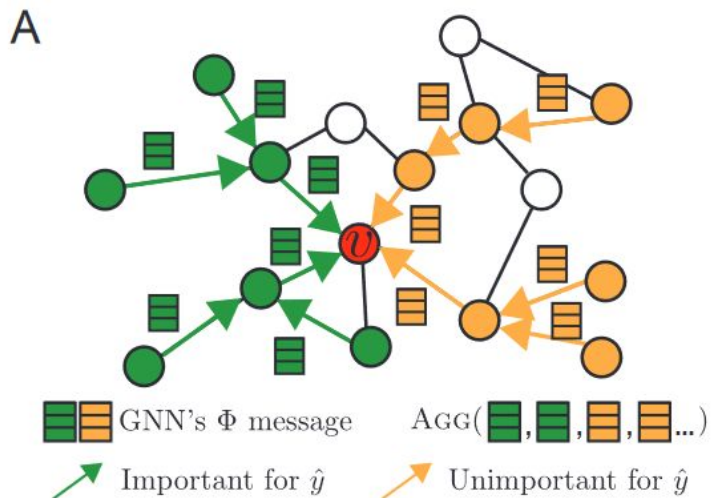
Assigning importance scores on edges



Assigning importance scores on connected subgraphs



- A: Edge masks for class-distinguish important edges
B: Node masks for important node feature dimension.



Explanation based on edges

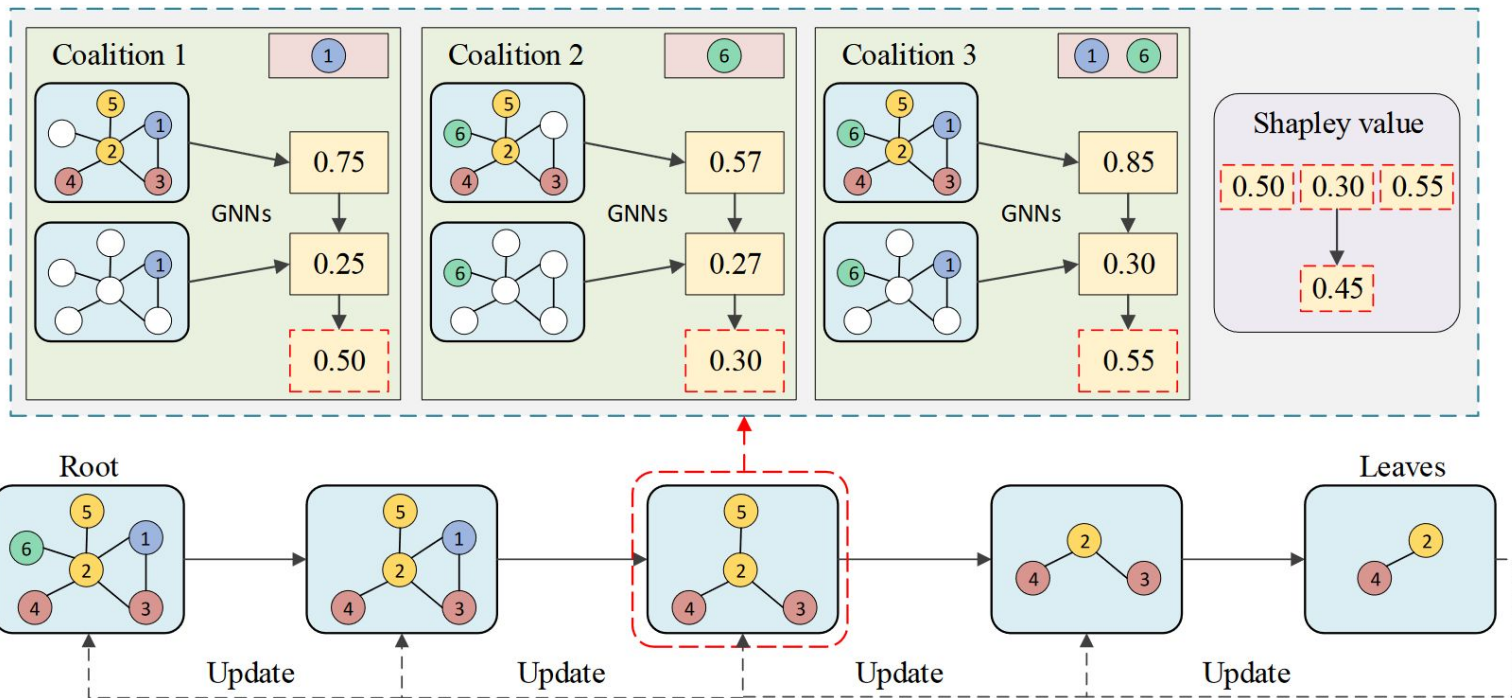
$$\min_M - \sum_{c=1}^C \mathbf{1}[y = c] \log P_{\Phi}(Y = y \mid G = A_c \odot \sigma(M_E), X = X_c \odot \sigma(M_X))$$

- It learns softmax masks on graph edges and node features.
- The masks are randomly initialized and updated to maximum the mutual information between the original predictions and the perturbed graphs.

SubgraphX



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Yuan et al. On Explainability of Graph Neural Networks via Subgraph Explorations

Department of Computer Science & Engineering

Explanation based on subgraphs

$$\mathcal{G}^* = \underset{|\mathcal{G}_i| \leq N_{\min}}{\operatorname{argmax}} \operatorname{Score}(f(\cdot), \mathcal{G}, \mathcal{G}_i)$$

- Explore the important subgraphs with Monte Carlo tree search algorithm
- Take Shapley value as the subgraph importance scores

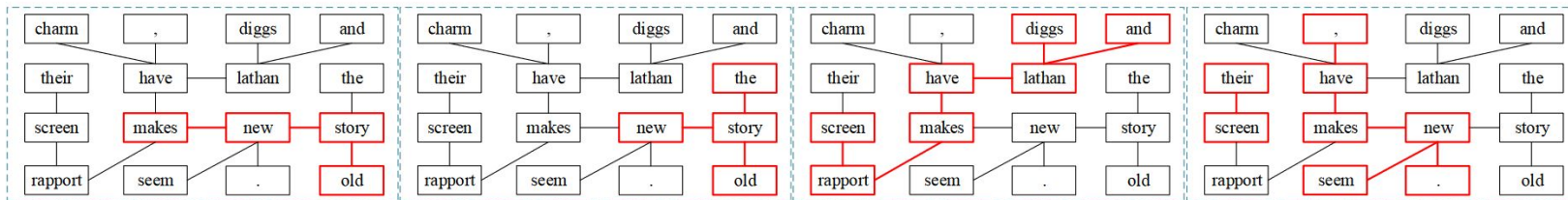
Sentiment Graph Dataset



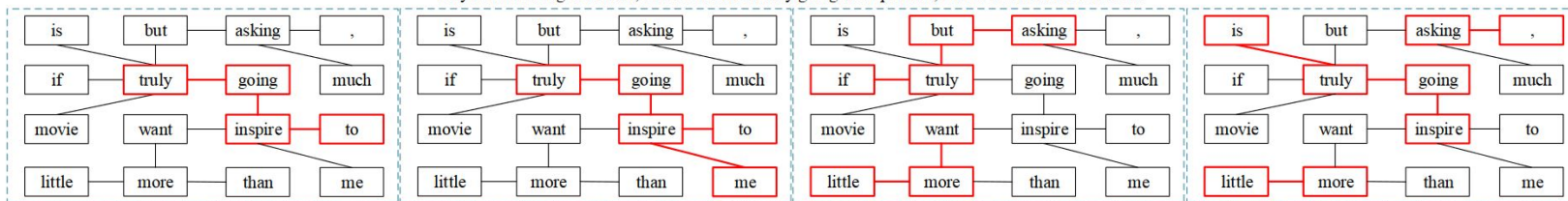
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- Better human evaluations for the explainability results
 - Graph-SST2, Graph-SST5, Graph-Twitter

“lathan and diggs have considerable personal charm, and their screen rapport makes the old story new.”



“maybe it is asking too much, but if a movie is truly going to inspire me, I want a little more than this.”



SubgraphX

MCTS_GNN

PGExplainer

GNNExplainer

Yuan et al. Explainability in Graph Neural Networks: A Taxonomic Survey

Yuan et al. On Explainability of Graph Neural Networks via Subgraph Explorations

Evaluation Metrics

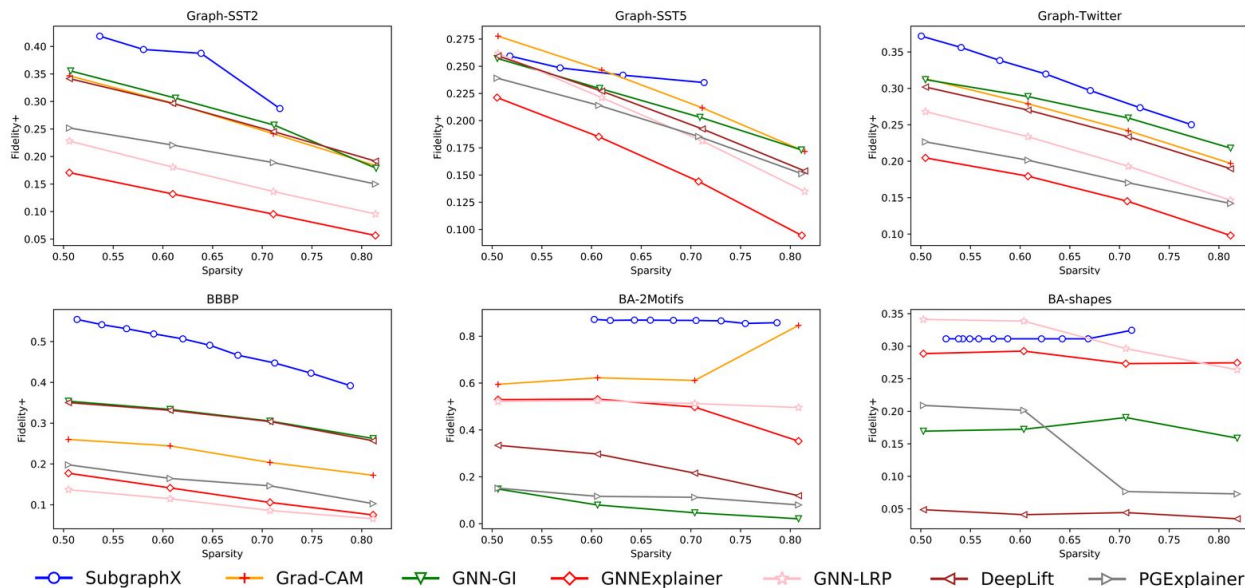


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- Compare different explainability methods systematically

- Fidelity+, Fidelity-, Sparsity

Benchmark of explainability methods using Fidelity and Sparsity



Yuan et al. Explainability in Graph Neural Networks: A Taxonomic Survey



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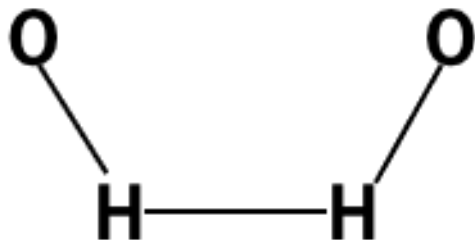
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Deep Learning on 3D Graphs

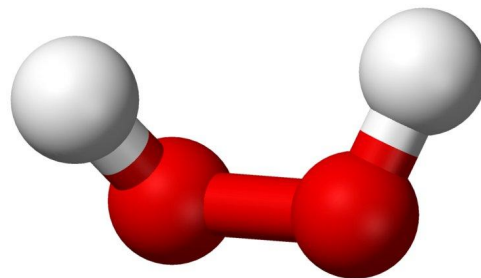
Molecules as Graphs



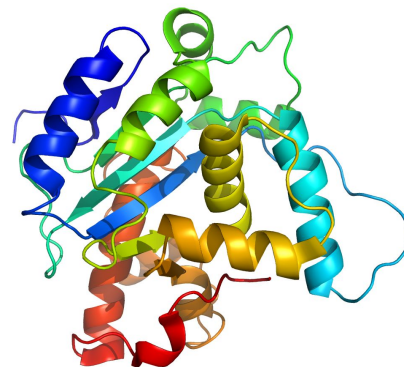
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Molecules as 2D
graphs



Position information:
Final conformation



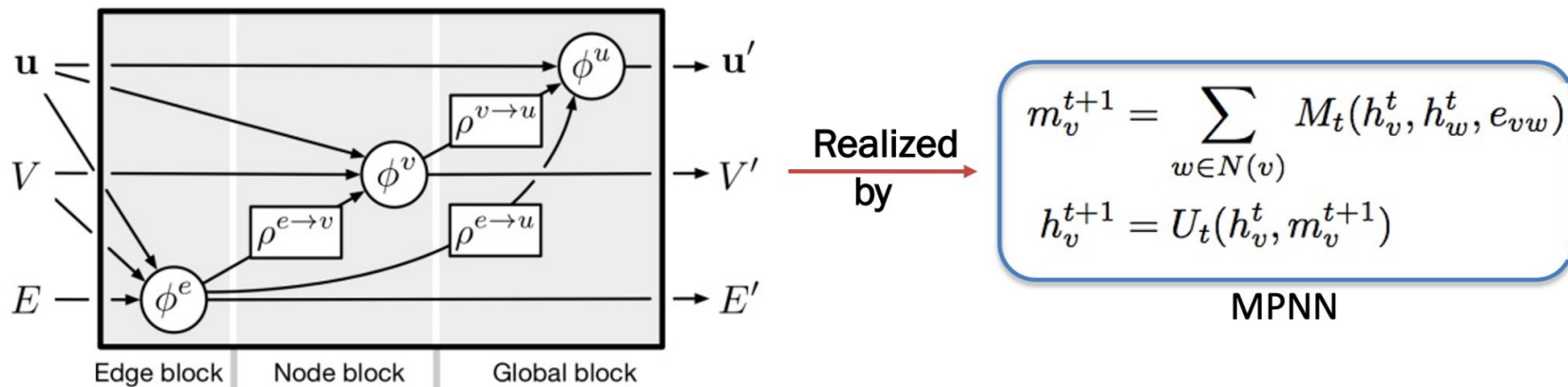
Macromolecules:
Protein

Figures are from <https://www.msdsolnline.com/2015/06/17/hydrogen-peroxide-health-safety-tips/>
<https://frontlinegenomics.com/alphafold-2-protein-structure-prediction-software-for-all/>

Current Neural Message Passing for Graphs



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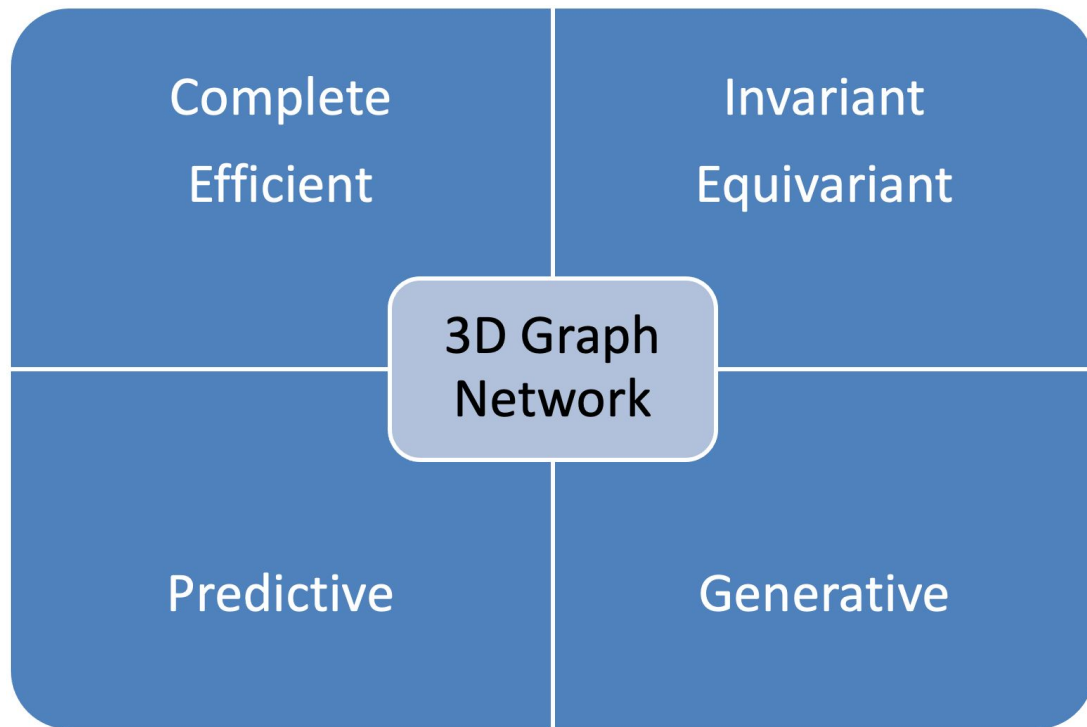
3D positional info is not captured!

Battaglia et al. "Relational inductive biases, deep learning, and graph networks." arXiv preprint arXiv:1806.01261 2018
Gilmer et al. "Neural message passing for quantum chemistry." ICML, 2017.

3D Graph Network



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- **Complete:**
 - Can fully capture 3D information?
- **Efficient:**
 - Complexity
- **Invariant/equivariant:**
 - To the rotation and translation of input 3D graphs
- **Prediction/generation**
- **Invariant, predictive**

Existing Methods for 3D Graphs



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- SchNet

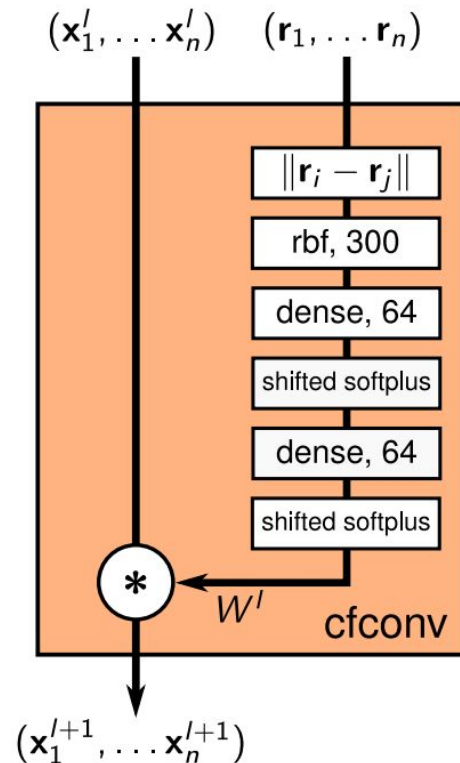
- Consider **distances between atoms**
- Message passing (continuous-filter convolutional layer):

$$\mathbf{x}_i^{l+1} = (X^l * W^l)_i = \sum_j \mathbf{x}_j^l \circ W^l(\mathbf{r}_i - \mathbf{r}_j)$$

- Distance embedding with radial basis functions:

$$d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$$

$$e_k(\mathbf{r}_i - \mathbf{r}_j) = \exp(-\gamma \|d_{ij} - \mu_k\|^2)$$



Schütt et al. "SchNet: A continuous-filter convolutional neural network for modeling quantum interactions." NeurIPS 2017.

- DimeNet

- Consider **distances between atoms and angles between edges**
- Message passing:

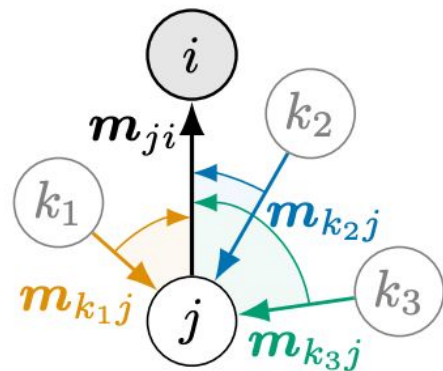
$$\mathbf{m}_{ji}^{(l+1)} = f_{\text{update}}(\mathbf{m}_{ji}^{(l)}, \sum_{k \in \mathcal{N}_j \setminus \{i\}} f_{\text{int}}(\mathbf{m}_{kj}^{(l)}, e_{\text{RBF}}^{(ji)}, \mathbf{a}_{\text{SBF}}^{(kj,ji)}))$$

- Distance embedding with radial basis function:

$$\tilde{e}_{\text{RBF},n}(d) = \sqrt{\frac{2}{c}} \frac{\sin(\frac{n\pi}{c}d)}{d}$$

- Distance and angle embedding with Fourier-Bessel basis:

$$\tilde{a}_{\text{SBF},ln}(d, \alpha) = \sqrt{\frac{2}{c^3 j_{l+1}^2(z_{ln})}} j_l(\frac{z_{ln}}{c}d) Y_l^0(\alpha)$$

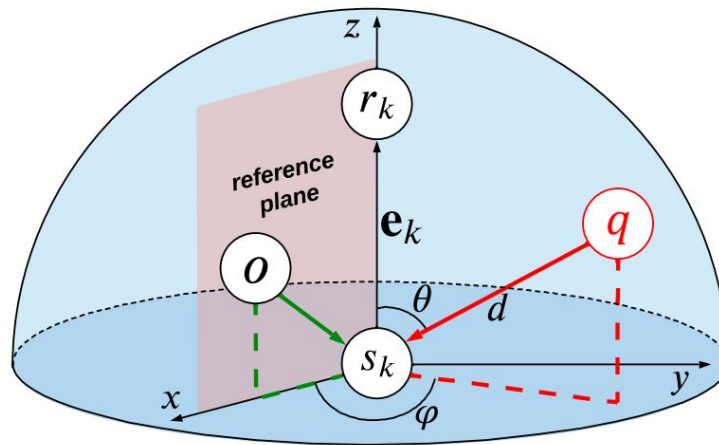
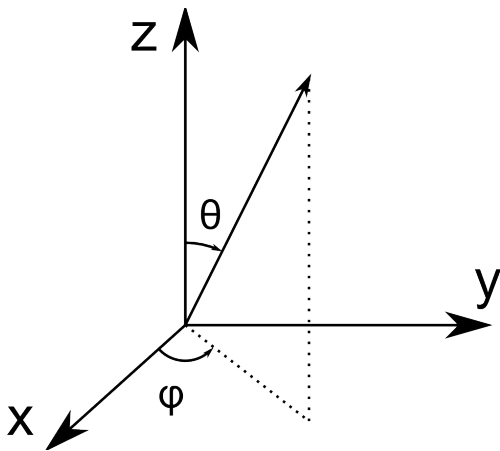


Existing Methods for 3D Graphs



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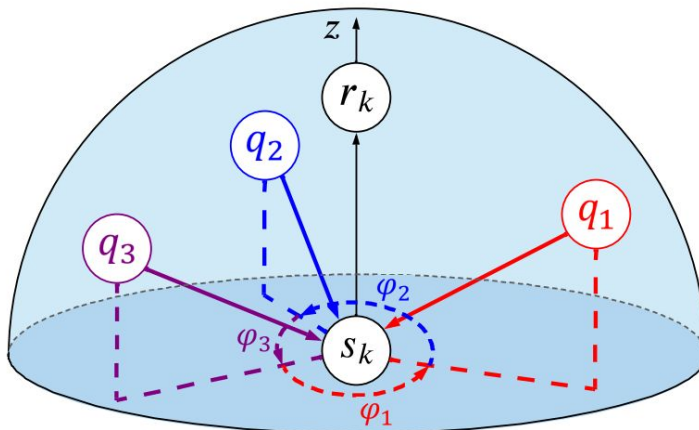
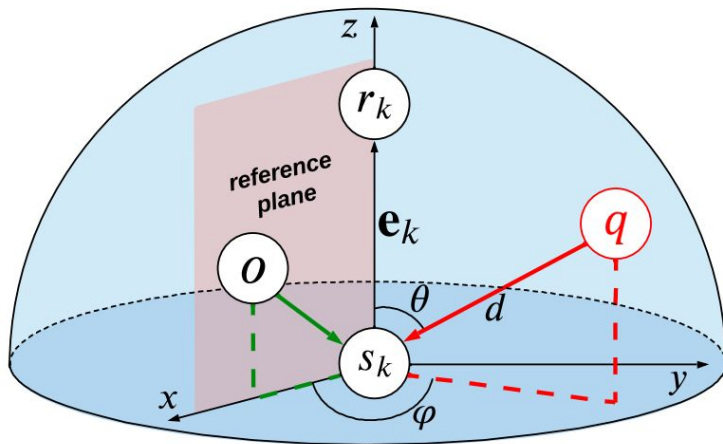
- SphereNet
 - Spherical coordinate



Liu et al. "Spherical message passing for 3d molecular graphs." ICLR. 2022.

- SphereNet
 - Spherical message passing

$$\mathbf{e}'_{s_k r_k} = f \left(\mathbf{e}_{s_k r_k}, \sum_{q \in N(s_k)} g(\mathbf{e}_{q s_k}, d_{q s_k}, \theta_{r_k s_k q}, \varphi_{r_k s_k q}) \right)$$



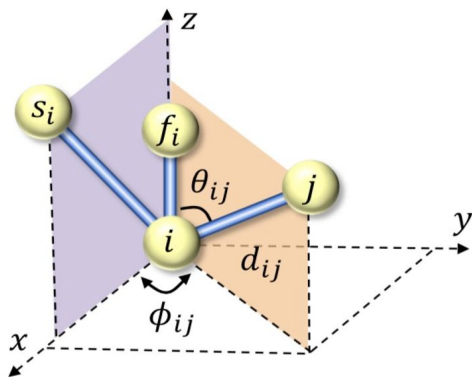
Liu et al. "Spherical message passing for 3d molecular graphs." ICLR. 2022.

Existing Methods for 3D Graphs



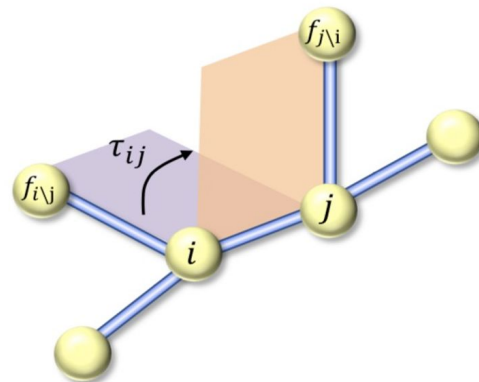
TEXAS A&M UNIVERSITY
Engineering

- ComENet
 - Complete representations of 3D structures
 - Local and global completeness



Local completeness:

- Spherical coordinates,
- New strategies to select reference nodes



Global completeness:

- Rotation angle of edge ij
- The only remaining degree of freedom

Wang et al. "ComENet: Towards Complete and Efficient Message Passing for 3D Molecular Graphs." arXiv preprint arXiv:2206.08515 (2022).

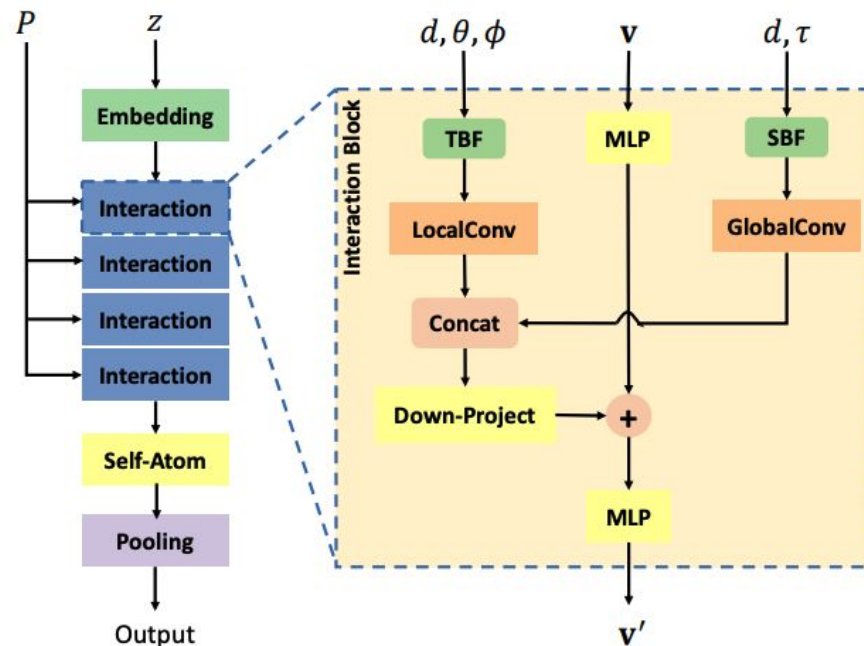
Existing Methods for 3D Graphs



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- ComENet
 - Message passing

$$\mathbf{v}'_i = g \left(\mathbf{v}_i, \sum_{j \in \mathcal{N}_i} f(\mathbf{v}_j, d_{ij}, \theta_{ij}, \phi_{ij}, \tau_{ij}) \right)$$



Wang et al. "ComENet: Towards Complete and Efficient Message Passing for 3D Molecular Graphs." arXiv preprint arXiv:2206.08515 (2022).

Existing Methods for 3D Graphs



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- Completeness and efficiency analysis

	Geometric information	Complete or not	Complexity	Message passing
SchNet	Distances d	×	$O(nk)$	Node update
DimeNet	Distances and angles (d, θ)	×	$O(nk^2)$	Edge update
SphereNet	Distances and angles (d, θ, ϕ)	×	$O(nk^2)$	Edge update
ComENet	Distances and angles (d, θ, ϕ, τ)	✓	$O(nk)$	Node update

Message passing: node update vs edge update

$$\mathbf{v}'_i = g\left(\mathbf{v}_i, f(\mathbf{v}_j, \mathbf{p}_i, \mathbf{p}_j)\right) \quad \mathbf{e}'_{ji} = g\left(\mathbf{e}_{ji}, f(\mathbf{e}_{kj}, \mathbf{p}_i, \mathbf{p}_j, \mathbf{p}_k)\right)$$

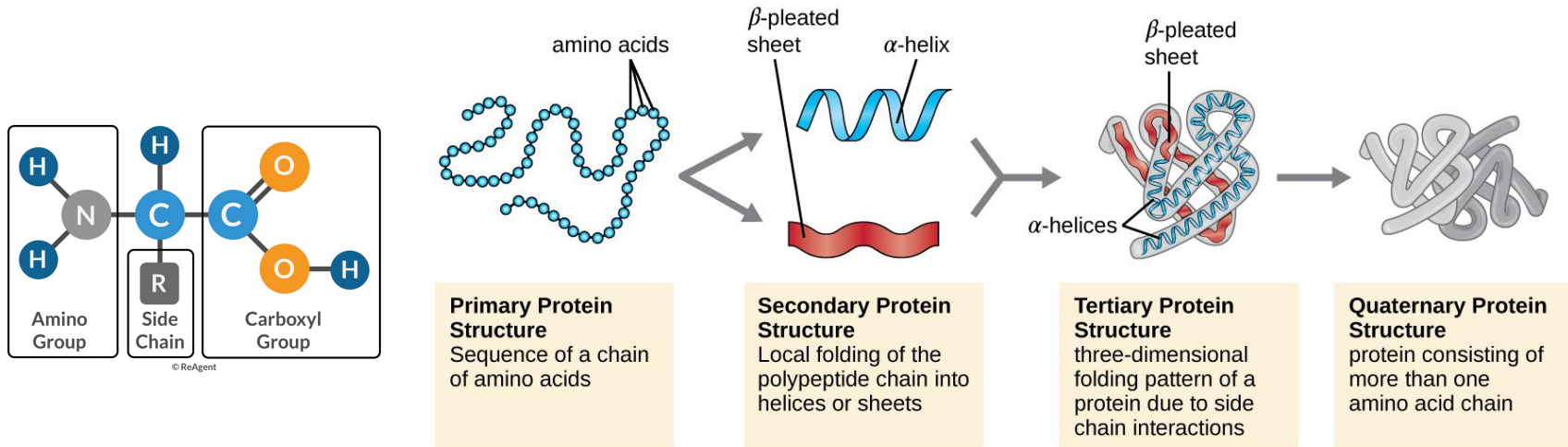
Complexity: n and k denote the number of nodes and the average degree in a 3D graph

Application on Proteins



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- Protein structure representation
 - Protein structure: *consist of amino acids*
 - Constructure protein graphs: *each amino acid as a node*
 - **Geometric information: *Ca coordinates, backbone structure, side chain structure***



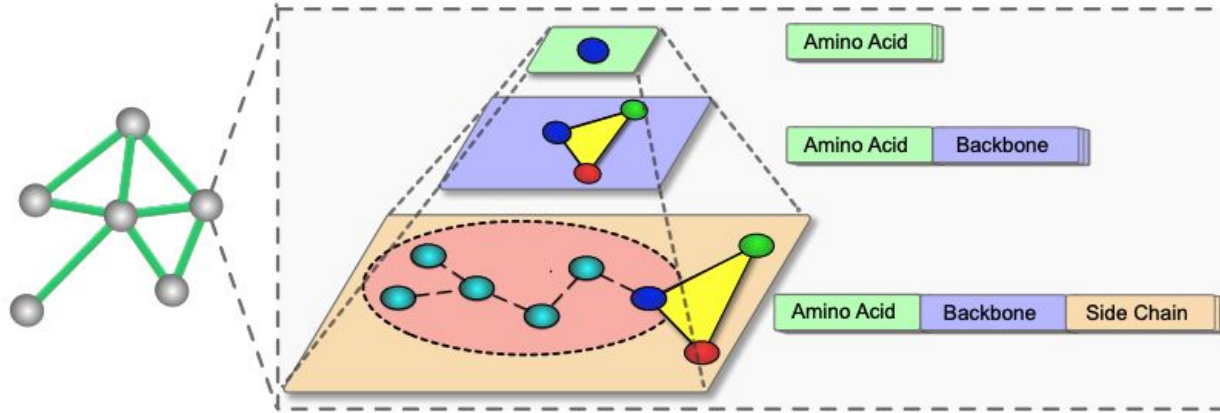
Figures are from <https://www.reagent.co.uk/what-are-amino-acids/>
[https://bio.libretexts.org/Bookshelves/Microbiology/Microbiology_\(OpenStax\)/07%3A_Microbial_Biochemistry/7.04%3A_Proteins](https://bio.libretexts.org/Bookshelves/Microbiology/Microbiology_(OpenStax)/07%3A_Microbial_Biochemistry/7.04%3A_Proteins)

Application on Proteins



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- ProNet
 - Geometric information: *Ca coordinates, backbone structure, side chain structure*



Wang et al. "Learning Protein Representations via Complete 3D Graph Networks." arXiv preprint arXiv:2207.12600 (2022).

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Q&A

